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"I wanna know what to expect" - Care needs regarding sexual and reproductive health after cancer in adolescence and young adulthood (AYA) and recommendations for providers

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ABSTRACT

Purpose: To assess care needs related to sexual and reproductive health of patients and survivors diagnosed with cancer during adolescence and young adulthood (AYA).
Methods: Participants ($N = 190$) were predominantly female (87.4%) and diagnosed with cancer between age 12–39 years. Common diagnoses included breast cancer (37.1%) or leukemia/lymphoma (27.2%). Cancer patients/survivors completed an online survey, which assessed care needs regarding sexual health and fertility/reproductive health. They provided written open answers, which were qualitatively analyzed.
Results: Care needs related to sexual health varied and included a need for more information about possible effects on sexual health, which will enable patients/survivors to better anticipate (physical and mental) side effects of cancer on sexual health. AYA patients/survivors need support from providers who normalize the topic of sex, who offer guidance in coping with numerous questions, who provide resources and practical tips and tricks, and who refer to specialists if needed. Participants would like providers to initiate conversations and communicate openly and honestly without taboo, bias, or shame. Care needs regarding reproductive health included needing support in navigating reproduction after cancer, while unmet information needs were central. This included needing information about fertility status and assessment options, reproduction/contraceptives, the safety of pregnancies, and the (im)possibility of alternatives to biological parenthood.
Conclusion: Communication is key in informing and supporting AYA cancer patients and survivors regarding their sexual and reproductive health. Providers should assess individual patients'/survivors' needs and tailor information that is tangible accordingly. Additional recommendations for healthcare providers are presented.

1. Introduction

Adolescence and young adulthood (AYA) is a life stage marked by various developmental tasks, such as exploring one's sexuality, engaging in romantic relationships, or potentially becoming a parent (Arnett, 2001, 2007). Being diagnosed with cancer during this life stage, together with treatment-related side effects, can have profound effects on sexual and reproductive health (see below), which remain neglected health domains in clinical care. Overall, supportive care remains suboptimal for AYAs with cancer (Baird et al., 2019; Janssen et al., 2021) and young survivors continue to have various unmet care needs (Perez et al., 2020b; Wong et al., 2017).

Sexual dysfunction is common in young cancer patients and survivors (Cherven et al., 2021; Zhou et al., 2015; Oveisi et al., 2023; Janssen et al., 2023), and associated worries or questions remain under-addressed in clinical practice (Lehmann et al., 2021; Wirtz et al., 2024; Cheung et al., 2024). This may be due to healthcare providers' hesitation, discomfort, lack of knowledge, bias, or lack of time to initiate

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conversations about sexual health (Albers et al., 2020b; Bober and Varela, 2012; Frederick et al., 2018; Katz, 2005; Perez et al., 2020a; Reese et al., 2017; Krouwel et al., 2015). Likewise, young patients/survivors can also be reluctant to discuss this topic with providers for various reasons, such as discomfort or the presence of a third party (e.g., family members) during consultations (Albers et al., 2020b; Canzona et al., 2019; Perez et al., 2020a; Reese et al., 2017). Consequently, up to 60% of survivors have unmet care needs related to sexual health (Lehmann et al., 2021) and initial insights into such needs indicate that young survivors need more information, support, and that providers' communication can be improved (Lehmann et al., 2021). Yet, the specific content or manners to provide such care remain unclear.

Relative to sexual health, more research attention has been paid to fertility preservation in young cancer patients over the last decade (Woodruff et al., 2021; Archer et al., 2024). Yet, providers' communication about infertility risks and fertility preservation at the time of diagnosis remains suboptimal (Wide et al., 2021; Quinn et al., 2020; Daly et al., 2019; Ehrbar et al., 2016; Srikanthan et al., 2019; Chapple et al., 2007; Schlossman et al., 2023; Yee et al., 2012; van den Berg et al., 2022; Kuntz et al., 2024). This can leave survivors feeling ill-informed and thinking that they miss crucial knowledge about their fertility later in life (Quinn et al., 2009; Perez et al., 2018; Kayiira et al., 2023; Zhang et al., 2020; Garvelink et al., 2015; Jardim et al., 2021). For example, young survivors miss information about how to assess their fertility status, which makes them feel uncertain about their reproductive health (Perz et al., 2014; Benedict et al., 2016, 2020b; Cherven et al., 2023; Eiser et al., 2011) and unable to possibly plan having a family accordingly (Halliday et al., 2014). Yet, what and how to provide fertility-related care in survivorship remains unclear.

Insights into care needs related to sexual and reproductive health are essential to offer guidance for healthcare providers to discuss these topics. Therefore, we (1) utilized the results of a systematic review (Lehmann et al., 2021) to examine supportive care needs regarding sexual health and we assessed (2) ongoing care needs regarding fertility and reproductive health in patients and survivors of AYA cancer.

2. Methods

2.1. Data collection

Data are part of the FROSA-study (FROSA = Fertility, Romance, and Sex after cancer in young Adulthood), which recruited participants through online convenience sampling on various social media/websites of patient organizations. FROSA targeted Dutch-speaking survivors who had been diagnosed with cancer between ages 18–39 years and who had completed treatment. Throughout recruitment (11/2022 – 04/2023), we were also approached by survivors diagnosed before age 18 and patients still undergoing treatment who also wanted to participate, which was not opposed given the expressed relevance of FROSA to these individuals. Participants provided informed consent online before they could start the survey anonymously. Our institution's medical ethical committee assessed this study as exempt from in-depth review (W21_282#21.309) and all procedures were in accordance with the Declaration of Helsinki.

2.2. Measures

Participants were presented with a brief overview of a literature review about *sexual* health-related care needs in young adult survivors (Lehmann et al., 2021), which had identified three needs: information, practical/emotional support, and communication needs. Participants were asked to elaborate and provide examples of each type of need. Moreover, participants were asked whether they had unmet care needs related to their *fertility* (yes/no). If endorsed, participants were asked to specify these support and information needs. Participants could type completely open answers (see Appendix-1).

2.3. Analyses

All written answers (separately for sexual and reproductive health) were imported into the software program MaxQDA and analyzed using content analysis (Hsieh and Shannon, 2005). Each participant's answer represented the smallest unit of data, which was inductively double-coded based on content. First, one coder (VL) familiarized themselves with one-third of all answers and developed an initial template of broad content-based categories that were repeatedly identified. Second, two coders (VL, BdO) independently coded the answers of $n = 25$ randomly selected participants utilizing this template and adjusted categories/labels as needed. Both coders then discussed their initial codes, resolved discrepancies, and continued to code answers of another $n = 25$ participants. Again, codes were discussed, discrepancies resolved, and both coders then coded all remaining answers. Throughout, similar codes were grouped together, further summarized to overarching themes, and discussed. Final overarching themes and descriptions are presented (see example quotes in Appendix-2/Appendix-3).

3. Results

3.1. Sexual health

$N = 190$ patients and survivors of AYA cancer completed the online survey. All were asked to elaborate on sexual health-related care needs, of which $n = 151$ participants provided at least one written answer. These $n = 151$ participants were predominantly female (87.4%), between 18 and 53 years old at study ($M = 32.8$ years), and diagnosed with cancer between age 12 and39 years. Time since diagnosis ranged between 0 and 19 years, and 6.6% were still on active treatment. The most common types of diagnoses included breast cancer (37.1%) or leukemia/lymphoma (27.2%; see details in Table 1).

Participants provided diverse descriptions of information, support, and communication needs related to sexual health. Information needs centered around needing more information and enabling anticipation. Support needs focused on guidance, which was often illustrated by examples of types of needed information to feel supported. Communication needs centered around participants' views of how providers should initiate and conduct conversations about sex (Table 2/Appendix-2).

3.1.1. Sexual health: Information needs

Although participant's answers varied, many voiced a need for **(1.) more information/knowledge transfer** about possible changes and side effects of cancer treatment on sexual health. This was further described as needing to (1.1) anticipate what to expect in general (i.e., raise awareness) by being given examples or told about stories of other young patients/survivors (to serve as source of information/illustration). Participants highlighted the importance of being able to (1.1.1) anticipate which physical side effects to expect, such as changes in sex drive, pain, physical discomfort, vaginal dryness, or hormonal changes. Yet, healthcare providers should also emphasize (1.1.2) the emotional effects of cancer on self-perception and mood, and cascading effects on sexual interest, sexual activities, and romantic relationships.

3.1.2. Sexual health: Support needs

A need for providers to **(2.) normalize the topic of sex** and patients' experiences was emphasized. This can provide patients with reassurance, recognition, and understanding. For example, simply hearing from providers that patients/survivors are "*not the only one*" can help them put their experiences into perspective. Moreover, a need for **(3.) guidance and offering solutions** was voiced to help patients and survivors cope with and navigate their possible sexual problems (e.g., knowing when to alert providers; times lines of recovering sex drive). Such guidance may also target numerous questions related to (3.1) "*How to deal with …*" which included needing help to cope with side effects, a

Table 1
Descriptive statistics of included participants for both parts: Sexual and reproductive health care needs.

	Sexual health-related care needs	Reproductive health-related care needs
	$N = 151$	$N = 71$
Age at study, M(SD), range	32.8 (6.2), 18-53	31.8 (5.9), 18-52
Sex		
Female	132 (87.4%)	64 (90.1%)
Male	19 (12.6%)	7 (9.9%)
Relationship status		
Partnered/married	109 (72.2%)	52 (73.2%)
Single	42 (27.8%)	19 (26.8%)
Sexual orientation		
Heterosexual	139 (92.1%)	67 (94.4%)
Homo-/bi-/pansexual/queer	10 (6.6%)	2 (2.8%)
missing	2 (1.3%)	2 (2.8%)
Age at 1st diagnosis		
12 or younger	2 (1.3%)[a]	2 (2.8%)[a]
13-17	6 (4.0%)[a]	1 (1.4%)[a]
18-24	32 (21.2%)[b]	13 (18.3%)[f]
25-29	43 (28.5%)[c]	24 (33.8%)[g]
30-34	47 (31.1%)[d]	23 (32.4%)[h]
35-39	21 (13.9%)[e]	8 (11.3%)
No relapse or second malignancy	125 (82.8%)	63 (88.7%)
Type of diagnosis (at AYA age)		
Breast cancer	56 (37.1%)	29 (40.8%)
Leukemia/lymphoma	41 (27.2%)	15 (21.1%)
Gynecological cancers	22 (14.6%)	14 (19.7%)
Testicular cancer	8 (5.3%)	2 (2.8%)
Sarcomas	8 (5.3%)	3 (4.2%)
other	16 (10.6%)	8 (11.3%)
Years since last diagnosis	3.6 (3.8), 0–19 years	3.6 (4.4), 0–24 years
Years since treatment completion		
On active treatment	10 (6.6%)	5 (7.0%)
< 1 year	59 (39.1%)	27 (38.0%)
1–5 years	54 (35.8%)	26 (36.6%)
≥ 5 years	26 (17.2%)	12 (16.9%)
Missing	2 (1.3%)	1 (1.4%)

[a] All relapsed/diagnosed with secondary malignancy between age 18-39.
[b] 6/32 relapsed/diagnosed with secondary malignancy between age 18-39.
[c] 7/43 relapsed/diagnosed with secondary malignancy between age 18–39 and another 2/43 relapsed/diagnosed with secondary malignancy after age 39.
[d] 2/47 relapsed/diagnosed with secondary malignancy between age 18-39.
[e] 1/21 relapsed/diagnosed with secondary malignancy after age 39.
[f] 3/10 relapsed/diagnosed with secondary malignancy between age 18-39.
[g] 1/24 relapsed/diagnosed with secondary malignancy between age 18-39.
[h] 1/23 relapsed/diagnosed with secondary malignancy between age 18-39.

changed body or self-perception, and feelings of unattractiveness (Table 2/Appendix-2). Other questions included how to regain their sex drive or a pleasurable sex life; and which role patients themselves can play in their recovery. Thereby, a need for (3.2) practical tips and tricks was frequently mentioned. This was sometimes specified by examples of needing aids/tools, such as lubricants, creams, or sex toys.

Participants also voiced a need for support in coping with romantic relationships. This included wanting to (3.3) learn how to communicate about cancer and sex with (potential) partners. Partnered participants expressed a need to learn more about how to deal with cancer and sex within their relationship and how to (re)create intimacy with a partner. Unpartnered participants described a need for help with dating. They also emphasized that more attention is needed for the sexual health issues of singles.

Patients and survivors of AYA cancer also need to be directed toward available and adequate **(4.) resources** of support, instead of "*having to google everything by yourself*." Moreover, a need for (4.1) fast and accessible referrals to sexologists or psychologists/therapists was frequently indicated. It was specified that anyone with relevant "expertise" would be adequate. Referrals to (4.2) peer support were frequently mentioned as another source of support, which were sometimes preferred to be anonymous. Several types of additional supporting (4.3) resources were suggested, predominantly including written information, which could either be provided on paper (e.g., folders) or online. The potentially added value of podcasts, video clips, online platforms/programs, as well as (in-person) information sessions or courses were mentioned.

Table 2
Care needs related to sexual health.

INFORMATION NEEDS	*n*
1. Information, and knowledge transfer (about possible changes/side effects)	56
1.1 know what to expect, raise awareness	10
- provide examples of what could happen; provide stories of others	20
1.1.1 **anticipate physical changes/problems**	**18**
- (reasons for) lower/changed sex drive	14
- vaginal dryness	12
- pain	3
- other, e.g. effects of (lower) hormone levels, menopause, recovery time, fertility)	11
1.1.2 **focus on physical and mental effects**	**21**
SUPPORT NEEDS [incl. examples of supportive information]	
2. Reassurance, normalization, recognition, understanding	34
3. Coping: needing support, guidance, and solutions: What is possible/allowed?	39
(e.g., coping with side effects (e.g., vaginal dryness), contraceptive options, Is sex safe/possible? other forms of intimacy; options for hormonal therapy; mental health support)	17
3.1 **How to … ?**	10
- cope/improve self-image, accept (changed) body, feel attractive	17
- regain/increase sex drive; have pleasurable sex; accept a lower sex drive	15
- other (e.g., … accept certain physical side effects, fatigue)	7
- contribute to own recovery	4
3.2 **(practical) tips & tricks; advice** (e.g., sex positions, touching)	21
- aids/tools (e.g., dilators, creams, toys)	7
- lubricants	8
3.3 **learn how to communicate with (potential) partners**	31
a How to deal with current partner/help experiencing intimacy	12
b Help dating/more awareness for unpartnered patients/survivors	6
4. Resources of support: Where to find help?	16
4. 1 **(fast) referral/access to:**	8
- sexologist	26
- 'specialists'/'experts'	16
- psychologist, (relation)therapist	12
- not own providers (oncologist/nurses), someone independent	9
- others (e.g., nurse specialist, social work, AYA team, physiotherapist)	14
4.2 **peer support**	30
4.3 **types of resources:**	
written information (folder/books *or* online)	22
podcasts, video clips/illustrations; online platforms; program, information sessions, courses	17
COMMUNICATION NEEDS	
5. Need for conversations/opportunity to talk	23
5.1 **provider initiates the conversation (or refers)**	20
standard part of care	9
5.2 **characteristics of communication:**	
- (pro)active, open, honest, direct, clear, accessible	49
- eliminating taboos; without having/creating shame	21
- empathic, comfortable, not too formal	7
- take patients seriously, provide space for a 'true' conversation, unbiased	19
- assess needs (e.g., ask questions, tailor to individual patient)	27
based on protocol/checklist/questionnaire *versus* open conversation	5
5.3 **conditions for a (good) conversation:**	
- trust, safe space; ensuring privacy	11
- being knowledgeable	4
- repeated conversations	13
- ensure sexual health as standard topic, e.g., through (follow-up) protocol	7
- provider characteristics: sex matched; not too old	4
5.4 **include romantic partners** (if existent)	16

3.1.3. Sexual health: communication needs

Overall, participants expressed a need for **(5.) conversations/the opportunity to talk**. Such conversations should be (5.1) initiated by providers, and patients should be referred if a provider does not feel qualified. Thus, it should be a standard part of care.

The (5.2) characteristics of such communication include that providers should be: *proactive, open, honest, direct, clear*, and *accessible*, while eliminating any *taboos*, and without creating any feelings of *shame/awkwardness*. Providers should be *empathetic* and *comfortable* talking about sex. They should give patients and survivors the feeling that they take them seriously (e.g., active listening, no computer typing) and provide space for a 'true' (e.g., interactive, not superficial) conversation, without rushing to conclusions; thus being unbiased. Providers should assess individual patients' needs and tailor their care to these needs (e.g., ask questions). Opinions differed about whether the assessment of patients' needs should be done during a conversation or based on written checklists/questionnaires.

The (5.3) preconditions for a (good) conversation, include creating trust and a safe space for patients/survivors, which also includes ensuring privacy (i.e., no third party present) and being knowledgeable. In addition, the topic of sex should be addressed repeatedly (before, during, and after treatment). Some participants suggested protocols for active and follow-up care to ensure that sexual health remains a standard topic in consultations. Very few participants also mentioned that providers themselves may not be too old or have the same sex as the patient to ensure an open conversation.

Finally, providers may also (5.4) involve romantic partners (if existent) in sexual health-related consultations. Providers should consider what partners may need to know, but they should have individual conversations with the patient first, before including a partner. Some participants also advised not to overemphasize partners' needs or to overemphasize penetrative sex (Table 2/Appendix-2).

3.2. Reproductive health

Of all $N = 190$ participants, 42.1% ($n = 80$) endorsed current unmet needs or questions regarding fertility, of which $n = 71$ provided answers to describe their needs (i.e., $n = 9$ provided unclear answers that were not coded). The $n = 71$ participants were predominantly female (90.1%), between 18 and 52 years old ($M = 31.8$ years), and diagnosed with cancer between age 12–39 years. Time since diagnosis ranged between 0 and 24 years, and 7.0% were still on active treatment. The most common types of diagnoses included breast cancer (40.8%) or leukemia/lymphoma (21.1%; Table 1).

Participants mainly voiced unmet **information** needs, which clustered into four topics: (a) fertility status and assessment, (b) reproduction/contraceptive use, (c) pregnancies, and (d) alternatives to biological parenthood (described below/Table 3). In light of these unmet information needs, participants also described a generic need for more and earlier support from providers. Participants elaborated that adequate and timely information can offer some peace of mind and more time to adjust to potential fertility problems. Greater awareness is needed for the grief over potential or confirmed fertility loss, even if someone already had children. At the same time, the topic of reproductive goals should not be forced into conversations nor should it be assumed that all patients/survivors want to have children (i.e., neglecting that reproductive goals entail a choice). Although only mentioned by a few participants, peer support may also be helpful in navigating fertility issues (e.g., discuss doubts about reproductive goals after cancer). Moreover, providers may pay support patients/survivors in navigating fertility-related discussions with (new) romantic partners or with existing young children (Table 3/Appendix-3).

3.2.1. Information needs: (a) fertility status and assessment

Survivors felt uncertain, wondered about their fertility, and wished to get more clarity. They needed information about the procedures of fertility assessments and where to access it. Moreover, information was needed about when to ideally get tested (i.e., how long after treatment completion), whether there is a time line for potential recovery of fertility, whether changes/developments in fertility status can be tested over time, and whether fertility assessments are covered by health insurance.

Female participants raised questions about when and what steps they should take if their menstrual cycle does not resume; when they might develop premature ovarian insufficiency (POI), and whether it is possible to delay the onset of POI (e.g., through hormone injections).

Table 3
Care needs related to reproductive health.

	n
Generic need for (earlier) SUPPORT/education; and creating more awareness/follow-up care	**14**
incl. possibly helping to find peer support; and support in communication (e.g., with new partners)	5
INFORMATION NEEDS:	
(a) **Fertility status and assessment: Am I still fertile?**	**17**
How does it work? When should I get tested? What's the recovery timeline? Financial aspects?	5
What should I do if my menstrual cycle does not return? When will I enter POI (can I delay it)?	4
(b) **Reproduction & Contraceptive Use**	
- When?	4
- natural conception *versus* assisted reproductive technologies (ART)	6
- ART: How does it work? How to utilize cryopreserved material	8
Does ART (IVF) increase chances of relapse?	1
- contraceptive use; suitable contraceptives during adjuvant hormone replacement therapy	2
(c) **Pregnancy**	**1**
c.1 risk for own health (e.g., risk of relapse due to pregnancy or pause adjuvant hormone replacement therapy?; risks to pregnancies during adjuvant hormone replacement therapy)	6
- options after hormone sensitive cancer?	2
- possible effects on pregnancies due to side effects, chemotherapy, and role of age	3
c.2 genetic & health risks for offspring	7
c.3 breastfeeding after cancer	2
(d) **Alternatives to biological parenthood**	8
- oocyte donation	3
- surrogacy (abroad); incl. emotional support[a]	3
- adoption/fostering, and its impossibility[a]	5

[a] Due to missing realistic options in the Netherlands (e.g., no commercial surrogacy; adoption/foster regulation).

3.2.2. Information needs: (b) reproduction & contraceptive use

Participants needed information about possible time lines of when it would be safe to try and conceive/sire a pregnancy following treatment; either naturally or through assisted reproductive technologies (ART). Thereby, various questions were indicated: For example, whether to choose natural conception or ART (e.g., whether one option would be superior or safer); and how long to attempt natural conception before transitioning to ART; or if there are ways to increase chances of conceiving. If having to utilize ART in the future, participants had additional questions about the procedures of ART.

Infertile participants raised questions about whether dismissing contraceptives would be safe, as they regarded contraceptives as superfluous in committed relationships. Other participants who were on adjuvant hormone replacement therapies (HRT) described dilemmas of little or no contraceptive options being available to them.

3.2.3. Information needs: (c) pregnancies

Female participants indicated questions about whether pregnancies increase their risk of cancer relapse. They also wondered whether pausing their adjuvant HRT (to get pregnant) would increase their risks of relapse; and whether there are health risks when becoming pregnant during adjuvant HRT. Women also questioned their odds for safe

pregnancies due to their advancing age, due to having had hormone-sensitive tumors, or due to certain treatment side effects (e.g., lymphedema).

Participants further needed information about genetic risks and health implications for a potential offspring. For example, they wondered whether oocytes are damaged by chemotherapy, whether it is safer to use cryopreserved materials (i.e., unexposed to cancer treatments), or if oocytes can generally contain cancer cells given that women are born with all their oocytes. Women also wondered if it is safe or possible to breastfeed after cancer.

3.2.4. Information needs: (d) alternatives to biological parenthood

Participants indicated having unanswered questions about which alternative family building options they may have if proven infertile. For example, they were interested in information about egg donation, including the procedures to find a donor, expected timelines, and associated costs. Other questions were about surrogacy (which likely has to be done abroad due to Dutch legislation), and whether adoption or fostering are even options to consider. Participants particularly needed to know how their cancer history may influence these options, and whether one could adopt as an unpartnered survivor. Thus, transparency about the obstacles or impossibility of alternative family building options was urgently needed.

4. Discussion

This study elicited more in-depth insights into care needs regarding sexual and reproductive health of patients and survivors of AYA cancer, given that previous research had established barriers and insufficiencies in talking about sexual and reproductive health in oncology (Lehmann et al., 2021; Wirtz et al., 2024; Albers et al., 2020b; Bober and Varela, 2012; Frederick et al., 2018; Katz, 2005; Perez et al., 2020a; Reese et al., 2017; Krouwel et al., 2015; Canzona et al., 2019; Daly et al., 2019; Schlossman et al., 2023; Quinn et al., 2009; van den Berg et al., 2022; Cheung et al., 2024). If discussed, AYA cancer patients/survivors tend to be dissatisfied with the provided information (Quinn et al., 2009; Perez et al., 2018; Kayiira et al., 2023; Zhang et al., 2020; Garvelink et al., 2015; Jardim et al., 2021) and experience unmet care needs (Perez et al., 2018, 2020b; Wong et al., 2017; Lehmann et al., 2021; Ehrbar et al., 2016; Yee et al., 2012; Garvelink et al., 2015). Participants in this study expressed that sexual and reproductive health should be a standard part of care. They described diverse topics and examples of unanswered questions and unmet needs, which can all serve as direct examples and guidance for clinical care. Together with our own clinical experiences, we summarized our recommendations in Tables 4 and 5.

4.1. Sexual health

Discussions about sexual health have been termed a two-way taboo where sometimes neither AYA cancer patients nor providers dare to initiate conversations (Graugaard et al., 2018). Nevertheless, due to various unmet needs, we argue that it is a provider's duty to at least initiate such conversation, preferably without third parties in the room. This will signal to patients and survivors that they can confide in their providers to find more information and support, if needed. The content or type of needed care depends on the individual AYA cancer patient/-survivor, which requires some type of needs assessment (e.g., during a conversation or based on a questionnaire) and tailoring care.

In essence, we identified that AYA cancer patients/survivors need information which enables them to anticipate possible effects of cancer and its treatment on sexual health. Thereby, an emphasis on both physical and mental effects of cancer and cascading effects on sexual activities and sexual health is important. Given that AYA cancer is relatively rare, patients and survivors often have little resources or stories from others, which likely also fuels their need to normalize the topic of sex. Reassuring AYA cancer patients/survivors that their questions and worries are valid can be highly beneficial. Please see all resulting recommendations in Table 4, while we also want to highlight here that if providers feel unprepared to have sex-related conversation, they need to refer patients to more specialized colleagues (i.e., stepped care approaches). Notably and depending on the issue/question, referrals to specialists (e.g., psychologists/sexologists) are not always needed. Instead, nurses or social workers may also serve as excellent resources, because oftentimes simple information (e.g., about lubricants) or guidance to (online) information can be sufficient (see Table 4).

Improved sexual health-related communication in (adult) oncology/hematology have been advocated for almost two decades (Schover et al., 2014; Katz, 2005; Ussher et al., 2020; Benedict et al., 2020a; Mick et al., 2003) and previous research even provided specific questions to help providers address this issue (Bober et al., 2016). However, predetermined questions or a checklist may not work well with AYA cancer patients/survivors, who prefer authenticity. Generally, the implementation of routine patient-reported outcome measures (PROMs) has been suggested as helpful to enhance patient-provider discussion in clinical care (Howell et al., 2015; Kotronoulas et al., 2014), without prolonging consultations (Graupner et al., 2021). Yet providers treating young patients should be mindful that if using PROMs, these should (a) be relevant to the AYA life stage, (b) be inclusive of diverse sexual orientations, gender identities, as well as singlehood and diverse forms of relationship formation, and (c) assess physical aspects of sexual functioning together with mental/emotional aspects of sexuality (e.g., orgasm function as well as pleasure). There are recent efforts to develop PROMs that assess (care) needs specific to AYA cancer patients/survivors (Patterson et al., 2022; Clinton-McHarg et al., 2012), but only one includes sexual health (Justafré et al., 2024).

Providers find it important that sexual health is discussed with young cancer patients (Albers et al., 2020b; Krouwel et al., 2020), but care remains suboptimal (Cheung et al., 2024), as evidenced by various unmet needs (Lehmann et al., 2021) and the current study. Strategies to improve providers' sexual health communication include role plays and structured training programs (Albers et al., 2020a), which improved providers' knowledge, but the implementation into routine clinical practice remains unclear (Albers et al., 2020a). Overall, we believe that it is crucial to include sexual health education and communication training into the curriculum of any healthcare profession. It starts with providers' willingness and readiness to talk about sex. In addition, assigning different roles to different providers of multidisciplinary care teams can be pivotal in ensuring that sexual health is discussed. For example, treating physicians (e.g., oncologists, hematologists, radiotherapists, or surgeons) may cover the medically-focused basis of sexual functioning, whereas nurses pay more attention to the intersection of physical and emotional aspects of sexual health. Moreover, further standardizing multidisciplinary care to include psychologists/sexologists who offer supportive care, which is also reimbursed by health insurance can further pave the way to more holistic approaches to oncological care (see all recommendations in Table 4). Notably, specialized care for AYA cancer patients has been implemented in the Netherlands since 2017. It aims to improve the above issues by educating providers about various age-specific issues and assigning a more prominent role to specialized AYA nurses (Jansen et al., 2018). The effects of such care on AYA cancer patients and survivors remain to be tested.

4.2. Reproductive health

Over the past decade, fertility preservation of young cancer patients has received more attention (Woodruff et al., 2021; Archer et al., 2024), and fertility-related concerns were commonly identified (Gorman et al., 2014, 2020). Although providers seem more comfortable talking about reproductive than sexual health, more than 40% of participants in this study reported ongoing unmet support and information needs. Thus, this

Table 4
Recommendations for providers: Sexual health.

1.The basics
- Awareness that sexual problems are prevalent in AYA oncology/hematology
- Knowledge of common sexual problems
- Knowledge of basic information/resources for patients (e.g., folders/websites (in different languages))
- No hesitation to talk about sex or knowledge which colleague is prepared to address sexual health
- Address the topic of sexual health with every patients, irrespective of background (e.g., relationship status) or clinical factors (e.g., type of diagnosis, stage of treatment)
- Prevent neglecting sex or assuming it is minor/unimportant. Unanswered questions or problems with sexual health can cause silent and unnecessary suffering
- Normalize the topic of sex for yourself

2.First step: Check-in
- Open the conversation naturally, e.g., when talking about related topics (e.g., reproductive goals, leisure time activities, side effects) and simply mention that patients/survivors can have sexual problems, concerns, or questions

AND:
- check if the patient/survivor can relate or is receptive to talking about sex
  - o If yes: try to understand the question(s)/problem(s). Ask for clarification if needed
    Note: Patients can sometimes be vague (e.g., "it feels different"), which needs clarification to prevent misunderstandings (e.g., What is "it"?, How is it "different"?)
  - o If not: reassure the patient/survivor that this is OK; and that they can come back to you if needed (maybe give them a folder or recommend a website)
    Note: It is not the intention to suggest patients/survivors should have problems, but to assess unmet needs. Thus, respect the patients' boundaries.
- Repeat the above check-in. One conversation will not be enough
- Remember: you do not have to have in-depth conversations about sexual health, but you (as part of a care team) should make sure that this topic is addressed by someone.

Note: The BETTER or PLEASURE model (Mick, 2007; Mick et al., 2003) might be helpful to realize the framework in which such first steps of a conversation can be helpful for patients

3.Next: A 'real' conversation

Patients and survivors of AYA cancer want open, honest, and authentic conversations, where they feel seen and heard. A rushed sex-related question that sounds like a checkbox on a long list of issues will not allow for a good conversation. Thus, a conversation with a nurse (practitioner) or social work might allow for a more in-depth conversation, than with a treating physician.
- Ensure privacy. If patients/survivors are older than 18 and third parties are in the room, send them out. Do not ask the patient/survivors if they are OK with a third party in the room. It is always more difficult for a patient to send them out vs. the provider telling third parties to leave the room, like a standard procedure.
- Be open, authentic, and curious:
  - O If a patient addresses issues that you are not familiar with (e.g., certain sexual preferences or side effects) stay curious and try to understand what the issue, question, or problem is. Only if you understand the problem, you might be able to help or refer to the right provider.
  - o If you don't know the answer: simply say so, but reassure the patient/survivor that you will try to find an answer
- Sometimes the only thing needed is listening and reassuring that other AYA cancer patients/survivors have similar questions/problems. Such reassurance and examples of others can be based on scientific/popular literature or providers'/colleagues' clinical experiences.
- Provide links/resources
- Refer to another specialist if indicated

study suggests that although infertility risks and fertility preservation are common topics at the time of diagnosis, fertility status and possible infertility are insufficiently discussed throughout survivorship. In line with previous research (Quinn et al., 2009; Perez et al., 2018; Kayiira et al., 2023; Zhang et al., 2020; Garvelink et al., 2015), participants in this study often lacked knowledge regarding fertility assessments in survivorship. We demonstrated additional unmet information needs regarding reproduction, healthy/safe pregnancies, and alternatives to biological parenthood, while emotional aspects of reproductive health (e.g., worries and concerns) tend to be neglected in clinical care (Glackin et al., 2024). Accordingly, some fertility-related information should be addressed more clearly and earlier in survivorship. For example, how, where, and when to complete fertility assessments or whether pregnancies or breastfeeding are contraindicated after cancer treatment. Likewise, clinical guidelines about the safety of pausing adjuvant HRT to conceive (Paluch-Shimon et al., 2022) may be discussed with patients to alleviate concerns. Regarding conceiving/family building options, information is difficult to find for AYA cancer patients/survivors, given that options and fertility status itself vary widely, depending on patients' cancer, treatment, and background factors. Thus, tailored information is essential (Schwab et al., 2020; Benedict et al., 2020a; van den Berg et al., 2022). Finally, alternatives to biological parenthood should be addressed with great care and applicable legislation should be consulted (see Table 5).

Table 5
Recommendations for Providers: Reproductive Health (in survivorship).

1.The basics
→ It is essential that providers in oncology, hematology, endocrinology, radiotherapy, surgery and/or reproductive medicine closely **collaborate** to provide valid and tailored information.

Ideally, hospitals will have trained oncofertility nurses that can act as navigator and source of information for patients, as done in some hospitals in the US (Smith et al., 2012; Whiteside et al., 2023; Vadaparampil et al., 2016). Similarly, specialized AYA-nurses, as utilized in the Netherlands, may also serve as navigators.
- Irrespective of having specialized nurses, know the referral system at your institute

2.Providing information & support
- Consider the individual patient/survivor and treatment history to tailor risk communication and fertility-related information
- Have clear conversations to provide a realistic outlook to patients/survivors on:
  - o Safety of pregnancies after cancer (for patients, partners, offspring)
  - o Reproductive options (unassisted vs. ART)
  - o Success rates and safety of ART
  - o (Im)probability of alternatives to biological parenthood (e.g., adoption, donations, surrogacy)
    - all alternatives are legally impossible in the Netherlands; check regulations in your country
    - address the potential emotional difficulties of such alternatives.
      Be careful in mentioning alternative family building options as easy or feasible options, because these options might offer hope, but can become a great disappointment for survivors
- Refrain from discussing fertility as a physical capacity only, but also:
  - o address related side effects, such as hormonal imbalances, menopausal symptoms, or bone health and discuss their implications for survivors (e.g., mood, fatigue, sexual interest)
  - o addresses patients'/survivors' goals and thoughts about fertility/family building
- Even if unintended, refrain from suggesting that having children is a universal life goal. Instead, assess patients/survivors' thoughts and attitudes.
- If patients/survivors experience grief over (potentially) becoming or being infertile, offer referrals to mental health professionals

During fertility-related conversations throughout treatment and survivorship, providers should also be mindful to prevent implicit assumptions, such that (possible) infertility is less intense for singles or patients who already had children. An open attitude is also important to not push the topic of reproductive goals and to emphasize its personal choice (i.e., some people never want to have children) throughout survivorship. Appendix-2 offers various examples of the content that may be addressed with AYA cancer patients/survivors, and together with our own clinical experience, we suggest the recommendations summarized in Table 5.

4.3. Limitations

Due to our online convenience sampling, we were unable to calculate response rates and selection bias may be present. We were also unable to ask follow-up questions due to our study design. Nevertheless, it allowed us to capture the voices of a rather large sample of AYA cancer patients/survivors. Our different approaches in assessing needs related to sexual health (based on a review) versus reproductive health (based on ongoing unmet needs) should be considered when interpreting our results (e.g., participants with reproductive health-related care needs in the past were not captured). Unfortunately, our sample did not include many males (i.e., >80% female) or individuals with diverse sexual identities (i.e., >90% heterosexual). These patients/survivors may be struggling with additional questions/topics. The multitude of reported examples and unanswered questions led us to assume that many more patients/survivors struggle in silence. We encourage future research to uncover more in-depth topics, problems, and possible changes in needs over time to further enrich the repertoire of issues that should be addressed in clinical care.

5. Conclusions

We hope to encourage various healthcare providers, who treat AYA cancer patients and survivors, to better address sexual and reproductive health, including psychosocial and emotional aspects (Tables 4 and 5). Notably, sexual and reproductive health may be addressed in conjunction (e.g., discuss sexual activities alongside contraceptive use or family planning). Education about sexual and reproductive health, together with communication training, should become routine parts of the curriculum of various medical, nursing, and psychosocial health professions; along with necessary financial resources for referrals to specialized care. Providers should assess needs, tailor information, and guide AYA cancer patients/survivors to available and suitable (online) resources. At the same time, patients and survivors need to be comfortable and ready to talk about sexual and reproductive health. Therefore, (online) resources may be of great help to better prepare them, while also always respecting their boundaries if they do not wish to talk about sexual or reproductive health.

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CRediT authorship contribution statement

Vicky Lehmann: Writing – review & editing, Writing – original draft, Validation, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Stephanie Both:** Writing – review & editing, Validation, Investigation. **Henk W. Elzevier:** Writing – review & editing, Validation, Investigation. **Jacqueline Tromp:** Writing – review & editing, Validation, Investigation. **Brenda den Oudsten:** Writing – review & editing, Investigation, Formal analysis.

Declaration of competing interest

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Appendix A. Supplementary data

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